

Task 1 — Document 02: Proposed Solution

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL) **Category:** Software | **Theme:** Smart Automation

1. Solution Overview

The **Sovereign AI Workbench** is a specialized, air-gapped, on-premise artificial intelligence workstation designed to serve as an intelligent digital assistant for MRPL industrial engineers.

By unifying open-weight large language models (LLMs), local visual text extraction (OCR), and dense vector similarity retrieval into a cohesive single-node FastAPI backend, the solution guarantees **complete data sovereignty**: zero bytes of data, prompts, embeddings, or inspection images ever cross the local network boundary.

High-Level Architecture Component Mapping

Core Component	Technical Subsystem	Primary Function
Local Multimodal OCR Pipeline	<code>glm-ocr:q8_0</code>	Extracts text from scanned images, nameplates, and maintenance logs.
Local RAG Vector Engine	<code>nomiic-embed-text</code> + Cosine Store	Ingests PDFs/TXTs, generates 768-dim embeddings, performs similarity search.
Specialized Model Task Router	Router Engine in <code>main.py</code>	Routes prompts to <code>qwen3.5:4b</code> (RAG/General) or <code>qwen2.5-coder:7b</code> (Coding).

2. Core Architectural Pillars

Pillar 1: Total Data Sovereignty & Offline Execution

- Built on top of **Ollama**, an open-source model execution runtime that hosts and executes GGUF/quantized models directly on local GPU/CPU hardware.
- Fully air-gap friendly. No outbound HTTP/HTTPS connections required after initial model deployment.

Pillar 2: Dense Retrieval-Augmented Generation (Local RAG)

- Incoming confidential documents (PDFs, TXT, Markdown) are automatically split into semantic text chunks.
- High-dimensional vector embeddings are generated using `nomiic-embed-text` locally.
- Chunks and embedding vectors are stored in a lightweight, local JSON index (`kb_index.json`).
- Semantic search computes cosine similarity between user queries and stored chunk embeddings to retrieve the top 5 most relevant context snippets in milliseconds.

Pillar 3: Grounded Answering & Strict Anti-Hallucination Guardrails

- When **Knowledge Mode** is enabled (`knowledge_enabled: true`), the LLM (`qwen3.5:4b`) is strictly instructed via system prompt engineering to answer **only** using the retrieved chunk context.
- If the semantic similarity score of retrieved context falls below threshold, or if the relevant information is absent from the Knowledge Base, the system responds:

"That information is not available in the knowledge base."

- Prevents dangerous speculative hallucinations during critical refinery equipment queries.

Pillar 4: Multimodal Inspection Image Parsing (GLM-OCR Integration)

- Plant engineers frequently capture physical equipment inspection logs, nameplates, or printed reports.
- Using `glm-ocr:q8_0`, the workbench automatically extracts structured text from `.png`, `.jpg`, `.jpeg`, and `.webp` images.
- Extracted OCR text is instantly chunked, embedded, and appended to the local knowledge base, making scanned reports instantly searchable alongside standard PDFs.

Pillar 5: Intelligent Task Routing & Dedicated Specialist Models

- Queries containing coding keywords (`python`, `code`, `fastapi`, `script`, `function`, `algorithm`, `sql`) are routed to `qwen2.5-coder:7b` for precise code synthesis.
- General operational or grounded document queries are handled by `qwen3.5:4b`.

3. Operational Workflow Matrix

User Input Type	Processing Pipeline	Model / Tool Used	Expected Output
PDF Document Upload	Text Extraction → Chunking → Local Vector Embeddings	<code>pypdf</code> + <code>nomic-embed-text</code>	Grounded KB Index Entry
Image / Scanned Log Upload	Image Extraction → Local OCR → Text Chunking → Embeddings	<code>glm-ocr:q8_0</code> + <code>nomic-embed-text</code>	Grounded KB Index Entry
Grounded Question (Knowledge ON)	Vector Cosine Search → Top-K Retrieval → Grounded Prompt	<code>nomic-embed-text</code> + <code>qwen3.5:4b</code>	Factual Answer + Source Citations
General Query (Knowledge OFF)	Direct Inference	<code>qwen3.5:4b</code>	General AI Response
Coding / Scripting Request	Keyword Task Router → Code Prompt Generation	<code>qwen2.5-coder:7b</code>	Executable Code / Script

4. Alignment with MRPL Industrial Safety & SOP Requirements

- 1. Refinery SOP Verification:** Allows maintenance crews to quickly verify exact step-by-step pump/compressor maintenance procedures directly from indexed MRPL manuals.
- 2. HSE (Health, Safety & Environment) Audit Readiness:** Instantly retrieves safety guidelines, hazardous material handling steps, and emergency shutdown procedures.
- 3. Equipment History Indexing:** Historical inspection reports (scanned or digital) can be uploaded in bulk, building an institutional memory accessible to all plant engineers.